

Measures of Dispersion

• Statistics:

→ Statistics is the collection, analyzation and representation of data.

* Measures of central tendency:
(Central Value or average)

① Arithmetic mean

→

> Individual series: $\bar{X} = \frac{\sum x}{n}$

> Continuous series / Discrete series:

$$\bar{X} = \frac{\sum fx}{N}$$

x = mid-value for continuous series.

> Assumed mean method:

$$\bar{X} = A + \frac{\sum fd}{N}; \text{ where } d = x - A$$

A = assumed mean

> Combined Mean:

$$\bar{X}_{12} = \frac{n_1 \bar{X}_1 + n_2 \bar{X}_2}{n_1 + n_2}$$

② Median

→ For individual and discrete series

$$Md = \left(\frac{N+1}{2} \right)^{th} \text{ term}$$

→ For continuous series

$$\text{Location of median} = \left(\frac{N}{2} \right)^{th} \text{ term}$$

$$Md = L + \frac{N/2 - C.f}{f} \times i$$

Where,

L = lower limit of median class

f = frequency

N = Total no. of observation = $\sum f$

$C.f$ = cumulative frequency of preceding class.

i = Step size

③ Mode

→ For individual and discrete series

Mo = Value with highest frequency.

→ For continuous series.

Location mode = class with highest frequency.

$$\text{Mode} = L + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times i$$

where,

L = lower limit of modal class

f_0 = frequency of preceding class

f_1 = frequency of modal class

f_2 = frequency of succeeding class

i = step size.

- Empirical relation:

$$\text{Mode} = 3 \text{ Median} - 2 \text{ Mean}$$

Measures of Dispersion:

→ ~~Dispersion~~ measures the degree of scatteredness of given data. Followings are the types of measures of dispersion:

① Absolute measures of dispersion

→ The measure of dispersion which is expressed in terms of original unit of given data is known as absolute measure of dispersion.

Some of them are:

> Range

> Quartile deviation (Q.D)

> Mean deviation (M.D)

> Standard deviation (S.D)

② Relative measures of dispersion

→ The measure of dispersion which is independent of unit of given data, is

known as relative measure of dispersion.

Some of them are:

- > Coe. of Range
- > Coe. of Q.D
- > Coe. of M.D
- > Coe. of S.D
- > Coe. of Variation (C.V)

Standard Deviation and Its coefficient

→ Root mean square deviation taken from mean

① Actual mean method / Direct method

→ ~~Actual mean method~~

$$\sigma = \sqrt{\frac{\sum f(x - \bar{x})^2}{N}}$$

② Shortcut Method

$$\sigma = \sqrt{\frac{\sum fx^2}{N} - \left(\frac{\sum fx}{N}\right)^2}$$

③ Assumed mean method (change of origin)

$$\sigma = \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N}\right)^2} \quad \text{Where } d = X - A$$

A = assumed mean

④ Step-deviation method (change of origin and scale):-

$$\sigma = \sqrt{\frac{\sum (fd')^2}{N} - \left(\frac{\sum f'd}{N}\right)^2} \times h \quad \text{Where } d' = \frac{x-A}{h}$$

A = assumed mean

h = scale factor

* Coeff. of standard deviation = $\frac{\sigma}{\bar{x}}$

* Coeff. of variation (C.V) = $\frac{\sigma}{\bar{x}} \times 100\%$

Note: The data with less value of C.V is considered to be more consistent.

: The square of standard deviation is called variance.

i.e. variance = σ^2

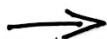
Exercise - 15.1

① Find the standard deviation of 20, 25, 10, 40, 50 by using

a) direct formula

- calculation of S.D

$$\sigma = \sqrt{\frac{\sum (x - (\bar{x}))^2}{N}}$$



x	$x - \bar{x}$	$(x - \bar{x})^2$
20	-9	81
25	-4	16
10	-19	361
40	11	121
50	21	441
$\Sigma x = 145$		$\Sigma (x - \bar{x})^2 = 1020$
$N = 5$		

Now,

$$\text{Mean } (\bar{x}) = \frac{\Sigma x}{N} = \frac{145}{5} = 29$$

$$\begin{aligned} \sigma &= \sqrt{\frac{\Sigma (x - (\bar{x}))^2}{N}} \\ &= \sqrt{\frac{1020}{5}} \\ &= \sqrt{204} \\ &= 14.28 \end{aligned}$$

b) change of origin

- Suppose $A = 25$ be the Assumed mean

$x \quad d = x - A \quad d^2$

20	-5	25
25	0	0
10	-15	225
40	15	225
50	25	625

$$\sigma = \sqrt{\frac{\Sigma d^2}{n} - \left(\frac{\Sigma d}{n}\right)^2}$$

$N = 5 \quad \Sigma d = 20 \quad \Sigma d^2 = 1100$

Now,

$$\sigma = \sqrt{\frac{\sum d^2}{n} - \left(\frac{\sum d}{n}\right)^2}$$

$$\sigma = \sqrt{\frac{1100}{5} - \left(\frac{20}{5}\right)^2}$$

$$\sigma = \sqrt{220 - 16}$$

$$\sigma = \sqrt{204}$$

$$\sigma = 14.28$$

From the answers of above two methods we can conclude that S.D is independent of change of origin.

5.

c) Find the standard deviation and coefficient of variation for the following distribution.

Marks	f	mid value(x)	fx	fx ²
10-20	5	15	75	1125
20-30	12	25	300	7500
30-40	15	35	525	18375
40-50	20	45	900	40500
50-60	10	55	550	30250
60-70	4	65	260	16900
70-80	2	75	150	11250
	N = 68		$\sum fx$ = 2760	$\sum fx^2$ = 125900

Now,

$$\begin{aligned}
 S.D &= \sqrt{\frac{\sum fx^2}{N} - \left(\frac{\sum fx}{N}\right)^2} \\
 &= \sqrt{\frac{125900}{68} - \left(\frac{2760}{68}\right)^2} \\
 &= \sqrt{1851.47 - \cancel{9} \cancel{1647.40}} \\
 &= \sqrt{204.06} \\
 &= 14.28
 \end{aligned}$$

And,

$$\bar{X} = \frac{\sum fx}{N} = \frac{2760}{68} = 40.58.$$

$$\begin{aligned}
 \therefore C.V &= \frac{S}{\bar{X}} \times 100\% = \frac{14.28}{40.58} \times 100\% \\
 &= 35.18\%
 \end{aligned}$$

b) Following are the marks obtained by two students A and B in 6 tests of 100 marks each.

Student A		Student B	
X	X ²	Y	Y ²
44	1936	48	2304
80	6400	75	5625
76	5776	54	2916
48	2304	60	3600
52	2704	63	3969
72	5184	69	4761

$$\Sigma x = 372 \quad \Sigma x^2 = 24304 \quad \Sigma y = 369 \quad \Sigma y^2 = 23175$$

No. of observation (n) = 6

$$(\bar{x}) = \frac{\Sigma x}{n} = \frac{372}{6} = 62$$

$$\begin{aligned} \sigma_x &= \sqrt{\frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2} \\ &= \sqrt{\frac{24304}{6} - \left(\frac{372}{6}\right)^2} \\ &= \sqrt{4050.66 - 3844} \\ &= \sqrt{206.66} \\ &= 14.37 \end{aligned}$$

Now,

$$\begin{aligned} C.V_x &= \frac{\sigma_x}{\bar{x}} \times 100\% \\ &= \frac{14.37}{62} \times 100\% \\ &= 23.18\% \end{aligned}$$

$$\frac{6}{12} = \frac{19}{19}$$

Again,

$$\bar{y} = \frac{\Sigma y}{n} = \frac{369}{6} = 61.5$$

$$\begin{aligned} \sigma_y &= \sqrt{\frac{\Sigma y^2}{n} - \left(\frac{\Sigma y}{n}\right)^2} \\ &= \sqrt{\frac{23175}{6} - \left(\frac{369}{6}\right)^2} \\ &= \sqrt{3862.5 - 3782.25} \\ &= \sqrt{80.25} \\ &= 8.95 \end{aligned}$$

Now,

$$\begin{aligned} C.V_y &= \frac{\sigma_y}{\bar{y}} \times 100\% \\ &= \frac{8.95}{61.5} \times 100\% = 14.55\% \end{aligned}$$

Here, Student B is more consistent
Thus, he should get the prize.

2) Find S.D., coe. of S.D and coefficient of variation in the following cases:

a) $n=10$, $\sum x = 55$, $\sum x^2 = 385$

→ Soln,

$$\begin{aligned} \text{S.D} &= \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2} \\ &= \sqrt{\frac{385}{10} - \left(\frac{55}{10}\right)^2} \\ &= \sqrt{\frac{385}{10} - \frac{121}{4}} \\ &= \sqrt{\frac{33}{4}} \end{aligned}$$

$$\sigma = 2.87$$

$$\therefore \text{S.D} = 2.87$$

Now,

Coeff. of S.D = ?

$$\bar{x} = \frac{\sum x}{n} = \frac{55}{10} = 5.5$$

$$\sigma = 2.87$$

$$\text{coe. of S.D} = \frac{\bar{x}}{\sigma}$$

$$= \frac{5.5}{2.87}$$

$$= 1.91$$

$$b) \sum f = 12, \sum fX = 108, \sum fX^2 = 1200$$

→ Soln,

$$\begin{aligned} S.D(\sigma) &= \sqrt{\frac{\sum fX^2}{N} - \left(\frac{\sum fX}{N}\right)^2} \\ &= \sqrt{\frac{1200}{12} - \left(\frac{108}{12}\right)^2} \quad \because \sum f = N = 12 \\ &= \sqrt{100 - 81} \\ &= \sqrt{19} \\ \sigma &= 4.35 \end{aligned}$$

Now,

$$\begin{aligned} \bar{X} &= \frac{\sum fX}{N} \\ &= \frac{108}{12} \\ &= 9 \end{aligned}$$

$$\begin{aligned} \text{Coe. of S.D} &= \frac{\sigma}{\bar{X}} \\ &= \frac{4.35}{9} \\ &= 0.48 \end{aligned}$$

$$\therefore \sigma = 4.35$$

$$\text{Coe. of S.D} = 0.48$$

3.

a) For a distribution the coefficient of variation is 22.5% and value of arithmetic mean is 7.5. Find out the value of standard deviation.

→ Soln,

$$\text{Coe. of Variation} = 22.5\%$$

$$\text{Arithmetic mean} = 7.5$$

$$\text{S.D} = ?$$

We know that,

$$\text{Coe. of Variation} = \frac{\sigma}{\bar{x}} \times 100\%$$

$$\Rightarrow 22.5\% = \frac{\sigma}{7.5} \times 100\%$$

$$\Rightarrow \frac{22.5}{100} = \frac{\sigma}{7.5}$$

$$\Rightarrow \sigma = \frac{22.5 \times 7.5}{100}$$

$$\Rightarrow \sigma = 1.6875$$

$$\therefore \text{Standard deviation} = 1.6875$$

b) coefficient of variation of two series are 75% and 90% and their standard deviations are 15 and 18 respectively. Find their mean.

→ Soln,

$$\text{coe. of Variation (C.V}_1) = 75\%$$

$$\text{coe. of Variation (C.V}_2) = 90\%$$

$$\sigma_1 = 15$$

$$\sigma_2 = 18$$

$$\text{means } (\bar{x}_1, \bar{x}_2) = ?$$

We know that,

$$\Rightarrow (\text{coe. of S.D}) = \frac{S.D(\sigma)}{\bar{X}} \times 100\%.$$

$$75 = \frac{15}{\bar{X}} \times 100\%.$$

$$\bar{X} = \frac{15 \times 100\%}{75}$$

$$\bar{X} = 20$$

Again,

$$\text{Coe. of S.D} = \frac{S.D(\sigma)}{\bar{X}} \times 100\%.$$

$$90 = \frac{18}{\bar{X}} \times 100\%.$$

$$\bar{X} = \frac{18 \times 100\%}{90}$$

$$\bar{X} = 20$$

$\therefore$ Means are 20 and 20.

- 4) The arithmetic mean of runs scored by three batsmen, A, B and C in the same series of 10 innings are 50, 48 and 42 respectively. The standard deviation of their runs are respectively 15, 12 and 7. If one of the three is to be selected on the basis of consistency, which player will be selected.

$\rightarrow$ Soln,

$$\bar{X}_A = 50$$

$$\sigma_A = 15$$

$$\bar{X}_B = 48$$

$$\sigma_B = 12$$

$$\bar{X}_C = 42$$

$$\sigma_C = 7$$

Now,

$$\begin{aligned} C.V_A &= \frac{\sigma_A}{\bar{x}_A} \times 100\% \\ &= \frac{15}{50} \times 100\% \\ &= 30\% \end{aligned}$$

$$\begin{aligned} C.V_B &= \frac{\sigma_B}{\bar{x}_B} \times 100\% \\ &= \frac{12}{48} \times 100\% \\ &= 25\% \end{aligned}$$

$$\begin{aligned} C.V_C &= \frac{\sigma_C}{\bar{x}_C} \times 100\% \\ &= \frac{7}{42} \times 100\% \\ &= 16.66\% \end{aligned}$$

Since coefficient of variance is low for student C, so he will be selected on the basis of consistency.

8. An analysis of monthly wages of workers of two organizations A and B yields the following result.

→ Soln,

	A	B
No. of workers	$n_1 = 40$	$n_2 = 50$
Avg wage	$\bar{x}_1 = 50$	$\bar{x}_2 = 60$
Variance	$\sigma_1^2 = 81$ $\Rightarrow \sigma_1 = 9$	$\sigma_2^2 = 100$ $\Rightarrow \sigma_2 = 10$

a) obtain the average monthly wages and standard deviation of wages of all workers in the two organization taken together.

→ We have,

$$\begin{aligned}\text{Combined avg wage } (\bar{x}_{12}) &= \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2} \\ &= \frac{40 \times 50 + 50 \times 60}{40 + 50} \\ &= \frac{2000 + 3000}{90} \\ &= \frac{5000}{90} \\ &= 55.56\end{aligned}$$

Again,

$$\text{Combined S.D } (\sigma_{12}) = \sqrt{\frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2}}$$

where,

$$\begin{aligned}d_1 &= \bar{x}_1 - \bar{x}_{12} = 50 - 55.56 \\ &= -5.56\end{aligned}$$

Similarly,

$$\begin{aligned}d_2 &= \bar{x}_2 - \bar{x}_{12} = 60 - 55.56 \\ &= 4.44\end{aligned}$$

Now,

$$\begin{aligned}\sigma_{12} &= \sqrt{\frac{40[9^2 + (-5.56)^2] + 50[10^2 + 4.44^2]}{40 + 50}} \\ &= \sqrt{\frac{40(111.91) + 50 \times 119.71}{90}} \\ &= \sqrt{\frac{4476.4 + 5985.68}{90}} \\ &= \sqrt{116.24} \\ &= 10.78\end{aligned}$$

b) which organization A or B has a large wage bill?

→ Soln,

$$\begin{aligned}\text{Wage bill of organization A} &= n_1 \bar{x}_1 \\ &= 40 \times 50 \\ &= 2000\end{aligned}$$

$$\begin{aligned}\text{Wage bill of organization B} &= n_2 \bar{x}_2 \\ &= 50 \times 60 \\ &= 3000\end{aligned}$$

∴ Organization B has large wage bill.

c) which organization is more equitable in regard to wages?

→ Soln,

For organization A,

$$\begin{aligned}C.V_A &= \frac{\sigma_1}{\bar{x}_1} \times 100\% \\ &= \frac{9}{50} \times 100\% \\ &= 18\%\end{aligned}$$

For organization B,

$$\begin{aligned}C.V_B &= \frac{\sigma_2}{\bar{x}_2} \times 100\% \\ &= \frac{10}{60} \times 100\% \\ &= 16.67\%\end{aligned}$$

$\therefore$ C.V_B is less than C.V_A so, Organization B is more equitable in regard to the wages.

9)

a) A group of 10 observations gave the arithmetic mean of 13 and a variance of 4. Later it was found that one observation 12 included in the data should have been 21. Find the correct mean and variance.

→ Soln,

$$* n = 10$$

$$\bar{X} = 13$$

$$\sigma^2 = 4$$

$$\Rightarrow \sigma = 2$$

Incorrect value = 12

Correct value = 21

$$\bar{X} = \frac{\sum x}{n}$$

$$13 = \frac{\sum x}{10}$$

$$\sum x = 130$$

Again,

$$S.D(\sigma) = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2}$$

$$\Rightarrow 2 = \sqrt{\frac{\sum x^2}{10} - \left(\frac{130}{10}\right)^2}$$

$$\Rightarrow 2 = \sqrt{\frac{\sum x^2}{10} - 169}$$

$$\Rightarrow 4 = \frac{\sum x^2}{10} - 169$$

$$\Rightarrow 1730 = \sum x^2$$

$$\therefore \sum x^2 = 1730$$

Now, the corrected values are as follows:-

$$\text{Correct } n = 10$$

$$\begin{aligned} \text{Correct } \sum x &= 130 - 12 + 21 \\ &= 139 \end{aligned}$$

$$\begin{aligned} \text{Correct } \sum x^2 &= 1730 - 12^2 + 21^2 \\ &= 1730 - 144 + 441 \\ &= 2027 \end{aligned}$$

Now,

$$\begin{aligned} \text{Correct } \bar{x} &= \frac{\text{Correct } \sum x}{\text{Correct } n} \\ &= \frac{139}{10} \\ &= 13.9 \end{aligned}$$

And,

$$\begin{aligned} \text{Variance (correct)} &= \frac{\text{Correct } \sum x^2}{\text{Correct } n} - \left(\frac{\text{Correct } \sum x}{n} \right)^2 \\ &= \frac{2027}{10} - 13.9^2 \\ &= 202.7 - 193.21 \\ &= 9.49 \end{aligned}$$

5.a) The following distribution is related with time spent by 54 visitors in using internet in a local cyber cafe.

Calculate S.D and Variance of time spent.

x	f	fx	fx ²
10	2	20	200
20	7	140	2800
30	15	450	13500
40	10	400	16000
50	20	1000	50000
N=54		$\Sigma fx = 2010$	$\Sigma fx^2 = 82500$

$$\bar{X} = \frac{\Sigma fx}{N} = \frac{2010}{54} = 37.22$$

Now,

$$\begin{aligned}
 \text{S.D } (\sigma) &= \sqrt{\frac{\Sigma fx^2}{N} - \left(\frac{\Sigma fx}{N}\right)^2} \\
 &= \sqrt{\frac{82500}{54} - (37.22)^2} \\
 &= \sqrt{1527.77 - 1385.32} \\
 &= \sqrt{142.4416} \\
 &= 11.93
 \end{aligned}$$

Again,

$$\begin{aligned}
 \text{Variance} &= \sigma^2 \\
 &= (11.93)^2 \\
 &= 142.44
 \end{aligned}$$

6. Two workers on the same job show the following results over a long period of time: Which worker ~~an~~ appears to be more consistent in time he requires to complete the job?

Worker A

Worker B

$$\bar{X}_1 = 30$$

$$\bar{X}_2 = 25$$

$$\sigma_1 = 6$$

$$\sigma_2 = 4$$

Now,

$$C.V._1 = \frac{\sigma_1}{\bar{X}_1} \times 100\%$$

$$= \frac{6}{30} \times 100\%$$

$$= 20\%$$

$$C.V._2 = \frac{\sigma_2}{\bar{X}_2} \times 100\%$$

$$= \frac{4}{25} \times 100\%$$

$$= 16\%$$

Since Coefficient of Variance of Worker B is less than A so Worker B is more consistent in time.

- 7) Find out the combined mean and standard deviation from the following data.

↳ Soln,

Series A

Series B

$$N_1 = 100$$

$$N_2 = 500$$

$$\bar{X}_1 = 50$$

$$\sigma_1 = 10$$

$$\bar{X}_2 = 60$$

$$\sigma_2 = 11$$

Now,

$$\bar{X}_{12} = \frac{n_1 \bar{X}_1 + n_2 \bar{X}_2}{n_1 + n_2}$$

$$= \frac{100 \times 50 + 500 \times 60}{100 + 500}$$

$$= \frac{5000 + 30000}{600}$$

$$= 58.33$$

$$= 58.33$$

$\therefore$ combined mean is 58.33

Again,

$$\text{Combined S.D.} = \sqrt{\frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2}}$$

Where,

$$d_1 = \bar{X}_1 - \bar{X}_{12} = 50 - 58.33 = -8.33$$

$$d_2 = \bar{X}_2 - \bar{X}_{12} = 60 - 58.33 = 1.67$$

Now,

$$\sigma_{12} = \sqrt{\frac{100(10^2 + (-8.33)^2) + 500(11^2 + (1.67)^2)}{100 + 500}}$$

$$= \sqrt{\frac{100 \times 169.38 + 500 \times 123.7889}{600}}$$

$$= \sqrt{\frac{16938.99 + 61894.45}{600}}$$

$$= \sqrt{131.38}$$

$$= 11.46$$

$\therefore$ Combined S.D = 11.46

9.b) The arithmetic mean and standard deviation of 19 observations are 40 and 6 respectively. At the time of checking it was found that one item 60 was missing. Find the new correct mean and standard deviation when the missing item is included. Also, find the correct value of coefficient of variation.

→ Soln,

$$N = 19$$

$$\bar{X} = 40$$

$$S.D(6) = 6$$

$$\bar{X} = \frac{\sum X}{N}$$

$$S.D = \sqrt{\frac{\sum X^2}{N} - \left(\frac{\sum X}{N}\right)^2}$$

$$40 = \frac{\sum X}{19}$$

$$6 = \sqrt{\frac{\sum X^2}{19} - 40^2}$$

$$\sum X = 760$$

$$36 = \frac{\sum X^2}{19} - 1600$$

$$\sum X^2 = 31084$$

Now,

The corrected values are as follow,

$$\text{Correct } N = 20$$

$$\text{Correct } \sum X = 760 + 60 = 820$$

$$\text{Correct } \sum X^2 = 31084 + (60)^2 = 34684$$

Then,

$$\text{Correct Mean } (\bar{X}') = \frac{\text{Correct } \sum X}{\text{Correct } N}$$

$$= \frac{820}{20}$$

$$= 41$$

$$\begin{aligned}
 \text{Correct S.D} &= \sqrt{\frac{C \cdot \Sigma X^2}{C \cdot N} - \left(\frac{C \cdot \Sigma X}{C \cdot N} \right)^2} \\
 &= \sqrt{\frac{34684}{20} - 41^2} \\
 &= \sqrt{1734.2 - 1681} \\
 &= \sqrt{53.2} \\
 &\approx 7.29
 \end{aligned}$$

Last,

$$\begin{aligned}
 \text{Correct Coe. of Variation} &= \frac{C \cdot S.D}{C \cdot \bar{X}} \times 100\% \\
 &= \frac{7.29}{41} \times 100\% \\
 &= 17.78\%
 \end{aligned}$$

$\therefore$ Correct Coe. of Variation = 17.78% $\downarrow$